Endurance exercise results in DNA damage as detected by the comet assay.

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To determine if 6 weeks of supplementation with antioxidants could alleviate exercise-induced DNA damage, we studied 21 runners during a 50 km ultramarathon. Subjects were randomly assigned to one of two groups: (1) placebos (PL) or (2) antioxidants (AO) (1000 mg vitamin C and 400 IU RRR-alpha-tocopheryl acetate). The comet assay was used to assess DNA damage in circulating leukocytes at selected time points: pre-, mid-, and 2 h postrace and daily for 6 days postrace. All subjects completed the race: run time 7.1 +/- 0.1 h, energy expenditure 5008 +/- 80 kcal for women (n = 10) and 6932 +/- 206 kcal for men (n = 11). Overall, the percentage DNA damage increased at midrace ($p < .02$), but returned to baseline by 2 h postrace, indicating that the exercise bout induced nonpersistent DNA damage. There was a gender x treatment x time interaction ($p < .01$). One day postrace, women taking AO had 62% less DNA damage than women taking PL ($p < .0008$). In contrast, there were no statistically significant differences between the two treatment groups of men at any time point. Thus, endurance exercise resulted in DNA damage as shown by the comet assay and AO seemed to enhance recovery in women but not in men.